

PU YANG

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About me

I am a PhD Student at the School of Mathematical Sciences, Peking University, where I work with Prof. Bin Dong. I am generally interested in inverse problem, reinforcement learning and large language model. My recent research interests are exploring the use of synthetic data in large language models.

Education

Peking University

Ph.D. major in Computational Mathematics

Beijing, China

Sep. 2021 – Present

Peking University

Undergraduate major in Mathematics

Beijing, China

Sep. 2017 – May. 2021

Publication

L2SR: Learning to Sample and Reconstruct for Accelerated MRI via Reinforcement Learning

with Bin Dong

- Published in Inverse Problem, 2024
- We propose an alternating training framework for jointly learning a good pair of samplers and reconstructors via deep reinforcement learning for accelerated MRI.

A Tale of Tails: Model Collapse as a Change of Scaling Laws

with Elvis Dohmatob, Yunzhen Feng, Francois Charton, and Julia Kempe

- Published in International Conference on Machine Learning (ICML 2024)
- We theoretically and empirically explore how the inclusion of synthetic data in training affects neural scaling laws in large language models, revealing potential risks of model collapse.

Beyond Model Collapse: Scaling Up with Synthesized Data Requires Reinforcement

with Yunzhen Feng, Elvis Dohmatob, Francois Charton, and Julia Kempe

- Accept by International Conference on Learning Representations (ICLR 2025)
- We theoretically and empirically explore how feedback-augmented synthesized data can mitigate model collapse in fine-tuning large language models.

MoColl: Agent-Based Specific and General Model Collaboration for Image Captioning

with Bin Dong

- Preprint
- We proposes a novel agent-enhanced model collaboration framework for complex image captioning tasks, designed to effectively integrate domain-specific and general knowledge.

Spend Wisely: Maximizing Post-Training Gains in Iterative Synthetic Data Bootstrapping

with Yunzhen Feng, Ciyuan Chen, Yuhang Wu, and Zhuoyuan Li

- Accept by Neural Information Processing Systems (NeurIPS 2025, Spotlight)
- We theoretically and empirically analysis how should the total budget on generation and training be allocated across iterations to maximize final performance, and demonstrate that an exponential growth policy exhibits significant advantages.

REAL-Prover: Retrieval Augmented Lean Prover for Mathematical Reasoning

with BICMR's AI4M Team

- Preprint
- We present REAL-Prover, a new open-source stepwise theorem prover for Lean 4, which consists of a fine-tuned large language model (REAL-Prover-v1) and an integrated with a retrieval system (Leansearch-PS), notably boosts performance on solving college-level mathematics problems.

Work experience

AI Algorithm Internship

In ByteDance

2021 Spring

Beijing, China

- I am primarily responsible for designing algorithms for audio fingerprinting, which involves extracting fingerprint features, searching through a speech library, and parallel acceleration, among other tasks. We utilize contrastive learning and propose a new feature extraction model that is more suitable for audio data, resulting in faster retrieval speeds and higher accuracy.

AI Algorithm Internship

In HiSilicon

2025 Spring to Summer

Shanghai, China

- I was involved in forward-looking research on next-generation LLM architectures. The core objective of my research was to enhance the accuracy and inference speed of long-context language models. My work centered on exploring innovative approaches, including dLLMs and sparse attention mechanisms, to address key challenges in scaling model performance.

Skills

Languages: Python, C/C++, MATLAB

Human Languages: Chinese, English

Development Tools: Pytorch, Linux (the administrator of our group)

Familiar Methods: computational mathematics, common machine learning methods, deep learning theory, reinforcement learning, large language model

Awards

Scholarships

in the undergraduate and doctoral program

- College Scholarship, School-level
- Outstanding Research Award, University-level

Modeling competition

in the undergraduate program

- MCM, Honorable Award

Olympiad in Informatics (OI)

in high school

- NOIP, First Prize

Mathematical Competition

in high school

- Peking University Summer Camp, Excellent camper (Admission to PKU)
- China Mathematics Competition, First Prize
- China Mathematics Olympiad, Silver Medal